

DOCUMENTO DE ABERTURA E ENCERRAMENTO

Ordem de Serviço - Laboratorio 04A

Mandatory Access Control (SELinux e AppArmor)

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Status: CONCLUIDA | SELinux Enforcing | Política default

1. Identificacao

Empresa	AmazonTech
Unidade	Gerencia de Infraestrutura e Seguranca da Informacao
Servidor	WEB-01 (MachadoPC) - Debian 13.6
Tipo	Auditoria, validacao e analise de mecanismos MAC
Data	06/08/2026
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato
Status da OS	CONCLUIDA

2. Situacao inicial x Tratamento

Situacao	Acao / Resultado	Status
MAC presente no ambiente	SELinux ativo no LSM do kernel; AppArmor analisado	Resolvido
Estado operacional	sestatus=enabled; getenforce=Enforcing	Validado
Politica carregada	Loaded policy name: default	Confirmado
Processos / contextos	ps -Z com system_u:system_r:kernel_t:s0	Identificado
Labels de arquivos	ls -Z: etc_t, shadow_t, sshd_exec_t, httpd_sys_content_t	Analisado
Ferramentas admin	sestatus, getenforce, ls -Z, ps -Z, restorecon, semanage	OK
Evidencias	Screenshots + saidas em lab04a/evidencias	Produzido

3. Evidencias principais

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Terminal - Evidencia 1 - Ambiente e LSM do kernel

root@MachadoPC:~# cat /etc/os-release | head -8
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"
NAME="Debian GNU/Linux"
VERSION_ID="13"
VERSION="13 (trixie)"
VERSION_CODENAME=trixie
DEBIAN_VERSION_FULL=13.6
ID=debian
HOME_URL="https://www.debian.org/"
root@MachadoPC:~# uname -a
Linux MachadoPC 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux
root@MachadoPC:~# cat /sys/kernel/security/lsm
capability,landlock,yama,safesetid,selinuxroot@MachadoPC:~# hostname; date
MachadoPC
Thu Aug 6 07:03:58 PM -04 2026
```

Figura - Evidencia 1 - Ambiente e LSM do kernel

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Terminal - Evidencia 2 - sestatus + getenforce (Enforcing)

root@MachadoPC:~# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:      /etc/selinux
Loaded policy name:          default
Current mode:                enforcing
Mode from config file:      enforcing
Policy MLS status:          enabled
Policy deny_unknown status: allowed
Memory protection checking:  actual (secure)
Max kernel policy version:   33

root@MachadoPC:~# getenforce
Enforcing

root@MachadoPC:~# grep -E '^(SELINUX|SELINUXTYPE)= ' /etc/selinux/config
SELINUX=enforcing
SELINUXTYPE=default

root@MachadoPC:~# # Coerencia sestatus x getenforce: ambos indicam Enforcing
root@MachadoPC:~# # Politica carregada: default (targeted/strict Debian)

```

Figura - Evidencia 2 - sestatus + getenforce (Enforcing)

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Terminal - Evidencia 3 - ls -Z (contextos SELinux)

root@MachadoPC:~# ls -lZ /etc/passwd /etc/shadow /usr/sbin/sshd /etc/hosts
-rw-r--r-- 1 root root system_u:object_r:net_conf_t:s0 415 Aug 6 19:03 /etc/hosts
-rw-r--r-- 1 root root system_u:object_r:etc_t:s0 1508 Jul 30 18:33 /etc/passwd
-rw-r--r-- 1 root shadow system_u:object_r:shadow_t:s0 1221 Jul 30 18:33 /etc/shadow
-rwxr-xr-x 1 root root system_u:object_r:sshd_exec_t:s0 568224 May 6 08:33 /usr/sbin/sshd

root@MachadoPC:~# ls -lZ / | head -16
total 2856
-rw-r--r-- 1 root root ? 5889 Jul 30 19:17 07_aide.txt
drwxr-xr-x 2 paulo auditoria system_u:object_r:default_t:s0 4096 Jul 28 19:00 auditoria
lrwxrwxrwx 1 root root system_u:object_r:bin_t:s0 7 May 12 2025 bin -> usr/bin
drwxr-xr-x 2 root root ? 4096 May 12 2025 boot
drwxr-xr-x 15 root root ? 3860 Aug 6 19:03 dev
drwxr-xr-x 79 root root system_u:object_r:etc_t:s0 4096 Aug 6 19:03 etc
drwxrwxr-x 2 maria financeiro system_u:object_r:default_t:s0 4096 Jul 28 19:00 financeiro
drwxr-xr-x 8 root root system_u:object_r:home_root_t:s0 4096 Jul 28 19:00 home
drwxr-xr-x 2 carlos infraestrutura system_u:object_r:default_t:s0 4096 Jul 28 19:00 infraestrutura
-rwxr-xr-x 1 root root ? 2836528 Apr 24 17:29 init
lrwxrwxrwx 1 root root ? 7 May 12 2025 lib -> usr/lib
lrwxrwxrwx 1 root root ? 9 May 12 2025 lib64 -> usr/lib64
drwx----- 2 root root ? 16384 May 12 18:45 lost+found
drwxr-xr-x 2 root root ? 4096 Aug 10 2025 media
drwxr-xr-x 8 root root ? 4096 Jul 23 16:10 mnt

root@MachadoPC:~# restorecon -Rv /var/www/html
Relabeled /var/www/html from <no context> to system_u:object_r:httpd_sys_content_t:s0
Relabeled /var/www/html/index.html from <no context> to system_u:object_r:httpd_sys_content_t:s0
root@MachadoPC:~# ls -lZ /var/www/html
total 4
-rw-r--r-- 1 root root system_u:object_r:httpd_sys_content_t:s0 18 Aug 6 19:03 index.html

```

Figura - Evidencia 3 - ls -Z (contextos SELinux)

```
Terminal - Evidencia 5 - restorecon (restauracao de labels)

root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:etc_t:s0 286 May 12 2025 /etc/motd
root@MachadoPC:~# chcon -t user_home_t /etc/motd
root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:user_home_t:s0 286 May 12 2025 /etc/motd
root@MachadoPC:~# restorecon -v /etc/motd
Relabeled /etc/motd from system_u:object_r:user_home_t:s0 to system_u:object_r:etc_t:s0
root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:etc_t:s0 286 May 12 2025 /etc/motd
```

Figura - Evidencia 5 - restorecon (restauracao de labels)

```
Terminal - Evidencia 6 - semanage fcontext (5843 regras)

root@MachadoPC:~# semanage fcontext -l | head -22
SELinux fcontext                                     type          Context
/                                                      directory     system_u:object_r:root_t:s0
/.*                                                    all files     system_u:object_r:default_t:s0
/\.ismount-test-file                                regular file   system_u:object_r:sosreport_tmp_t:s0
/\.journal                                            all files     <<None>>
/a?quota\.(user|group)                              regular file   system_u:object_r:quota_db_t:s0
/afs                                                  directory     system_u:object_r:mnt_t:s0
/bacula(/.*)?                                        all files     system_u:object_r:bacula_store_t:s0
/boot                                                  directory     system_u:object_r:boot_t:s0
/boot/.*                                              all files     system_u:object_r:boot_t:s0
/boot/System\.map(-.*)?                              regular file   system_u:object_r:system_map_t:s0
/boot/\.journal                                       all files     <<None>>
/boot/a?quota\.(user|group)                          regular file   system_u:object_r:quota_db_t:s0
/boot/efi(/.*)?/System\.map(-.*)?                    regular file   system_u:object_r:system_map_t:s0
/boot/lost\+found                                    directory     system_u:object_r:lost_found_t:s0
/boot/lost\+found/.*                                all files     <<None>>
/cgroup                                               directory     system_u:object_r:cgroup_t:s0
/cgroup/.*                                           all files     <<None>>
/dev                                                  directory     system_u:object_r:device_t:s0
/dev/(misc/)?agpgart                                character device system_u:object_r:agp_device_t:s0
/dev/(misc/)?psaux                                  character device system_u:object_r:mouse_device_t:s0

... total de regras fcontext: 5843

root@MachadoPC:~# semanage fcontext -l | grep -E 'httpd|sshd|www' | head -12
/etc/WebCalendar(/.*)?                                all files     system_u:object_r:httpd_sys_rw_content_t:s0
/etc/apache(2)?(/.*)?                                all files     system_u:object_r:httpd_config_t:s0
/etc/apache-ssl(2)?(/.*)?                            all files     system_u:object_r:httpd_config_t:s0
/etc/cherokee(/.*)?                                  all files     system_u:object_r:httpd_config_t:s0
/etc/drupal.*                                          all files     system_u:object_r:httpd_sys_rw_content_t:s0
/etc/glpi(/.*)?                                       all files     system_u:object_r:httpd_sys_rw_content_t:s0
... (saida truncada) ...
```

Figura - Evidencia 6 - semanage fcontext (5843 regras)

4. Conformidade com a Política AmazonTech

Diretriz	Atendida?
1. Servidor com MAC quando suportado pela distro/kernel	SIM
2. MAC complementa permissões tradicionais (DAC)	SIM
3. Alterações de política avaliadas/registradas	SIM
4. Estado operacional validado antes de produção	SIM
5. Processos críticos sob política MAC	SIM
6. Verificações com evidências	SIM
7. Nenhuma configuração alterada sem validação	SIM
8. Observações da configuração registradas	SIM
9. MAC integra política permanente de Hardening	SIM
10. Conclusões subsidiam Lab 4B	SIM
Resultado	OS CONCLUÍDA - camada MAC (SELinux Enforcing) validada no WEB-01
Documento técnico	Lab04A_Mandatory_Access_Control_PREENCHIDO.pdf
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato